

Middle East Technical University

Department of Electrical and Electronics Engineering

EE568 – Design of Electrical Machines

Project 2: Motor Winding Design & Analysis

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1 Introduction

This report presents the winding design and analytical modelling of a three-phase surface-mount permanent-magnet synchronous machine (SPMSM). The assigned design is **Design 1** (student ID 2575421, $2575421 \bmod 5 = 1$), with $N_s = 15$ stator slots and $N_m = 8$ rotor poles ($p = 4$ pole-pairs).

The report is organised as follows. Section 2 provides the theoretical background from standard references. Section 3 covers integral-slot winding design for a 72-slot, 6-pole machine. Section 4 presents the fractional-slot winding analysis for Design 1. Section 5 contains all analytical modelling calculations. Section 6 will present the 2D FEA results from ANSYS Maxwell.

2 Theoretical Background

2.1 Winding Fundamentals

An AC machine winding converts an applied current distribution into a rotating magnetomotive force (MMF) and, in the generating case, converts a rotating flux into an induced electromotive force (EMF). The quality of this conversion is quantified by the *winding factor* k_w , which accounts for the spatial distribution of conductors [1, 2].

2.1.1 MMF Distribution and Harmonics

For a coil placed in slots the MMF is not sinusoidal but contains space harmonics whose order is $n = 1, 5, 7, 11, \dots$ (for a balanced three-phase winding with no triplen harmonics). The n -th space harmonic amplitude is reduced relative to the fundamental by the *harmonic winding factor* $k_{w,n}$ [1].

Suppressing high-order space harmonics is desirable because they increase iron losses, create parasitic torque pulsations, and generate acoustic noise.

2.2 Integral-Slot Distributed Winding

An integral-slot winding has an integer number of slots per pole per phase:

$$q = \frac{N_s}{2pm} \in \mathbb{Z}^+ \quad (1)$$

For the 72-slot, 6-pole, three-phase machine of Q1: $q = 72/(6 \times 3) = 4$.

2.2.1 Distribution Factor

When q coils are placed in adjacent slots separated by one slot pitch α_e (electrical), their phasors are offset and the resultant EMF is smaller than the arithmetic sum. The ratio

is the distribution factor [1, Ch. 4]:

$$k_{d,n} = \frac{\sin(n q \alpha_e / 2)}{q \sin(n \alpha_e / 2)}, \quad \alpha_e = \frac{p \cdot 360^\circ}{N_s} \quad (2)$$

For the fundamental ($n = 1$) a larger q brings $k_{d,1}$ closer to unity and simultaneously reduces higher harmonics.

2.2.2 Pitch Factor

A full-pitched coil has its sides exactly one pole pitch τ_p apart. Shortening the coil to $y < \tau_p$ introduces a phase shift between the two sides, yielding the pitch factor [1, Ch. 4]:

$$k_{p,n} = \sin\left(\frac{n y \pi}{2 \tau_p}\right) \quad (3)$$

Short-pitching by $1/n$ of the pole pitch eliminates the n -th harmonic entirely ($k_{p,n} = 0$). The 11/12 short pitch ($y = 11$, $\tau_p = 12$) is chosen to suppress the 11th and 13th harmonics in a three-phase machine [3].

2.2.3 Winding Factor

The overall winding factor is:

$$k_{w,n} = k_{d,n} \cdot k_{p,n} \quad (4)$$

A high $k_{w,1}$ maximises torque density; a low $k_{w,n}$ for $n > 1$ reduces harmonic content.

2.3 Fractional-Slot Concentrated Winding (FSCW)

In a fractional-slot winding $q \notin \mathbb{Z}^+$, meaning the number of slots per pole is not an integer. When $q < 1$ and each coil is wound around a single tooth (coil pitch = 1 slot), the winding is called a *concentrated* or *tooth-coil* winding. This topology offers short end-windings, high slot fill factors, and easy manufacture, at the cost of lower fundamental winding factor and richer harmonic spectrum compared to a distributed winding [2, Ch. 4].

2.3.1 Star-of-Slots Method

For a fractional-slot winding the winding factor cannot be computed from simple closed-form expressions; instead the *star of slots* (phasor diagram) method is used [1, 4].

Each slot k is assigned an EMF phasor at electrical angle:

$$\varphi_k = (k - 1) \alpha_s \pmod{360^\circ}, \quad \alpha_s = \frac{N_m \cdot 180^\circ}{N_s} \quad (5)$$

The N_s phasors form the *star of slots*. For a three-phase winding the circle is divided into six 60° sectors and each phasor is assigned to the phase whose axis is nearest:

$$\begin{aligned} \text{A+}: [0^\circ, 60^\circ); \quad \text{C-}: [60^\circ, 120^\circ); \quad \text{B+}: [120^\circ, 180^\circ); \quad \text{A-}: [180^\circ, 240^\circ); \quad \text{C+}: \\ [240^\circ, 300^\circ); \quad \text{B-}: [300^\circ, 360^\circ) \end{aligned}$$

The n -th harmonic winding factor is then [1]:

$$k_{w,n} = \frac{1}{N_{ph}} \left| \sum_{k \in \text{ph}^+} e^{jn\varphi_k} + \sum_{k \in \text{ph}^-} e^{j(n\varphi_k + \pi)} \right| \quad (6)$$

where N_{ph} is the number of slots per phase, and ph^+ , ph^- are the positive and negative slot sets of that phase.

2.4 Analytical Machine Modelling

2.4.1 Air-Gap Flux Density – Surface PM Machine

For a surface-mount PM machine with radially magnetised magnets, treating the rotor surface and stator bore as concentric cylinders, the peak air-gap flux density is [2, Ch. 3]:

$$B_{g,\text{peak}} = \frac{B_r l_m}{l_m + \mu_r k_c g} \quad (7)$$

where B_r is the remanence, l_m the magnet radial thickness, μ_r the magnet relative permeability, g the physical air gap, and k_c Carter's coefficient.

2.4.2 Carter's Coefficient

Stator slotting increases the effective air gap. Carter's coefficient accounts for this by replacing the slotted stator with an equivalent smooth surface [5, 1]:

$$k_c = \frac{\tau_s}{\tau_s - \gamma}, \quad \gamma = \frac{4g}{\pi} \left[\frac{w_s}{2g} \arctan \frac{w_s}{2g} - \ln \sqrt{1 + \left(\frac{w_s}{2g} \right)^2} \right] \quad (8)$$

where $\tau_s = 2\pi R_{si}/N_s$ is the slot pitch at the stator bore and w_s is the slot opening width.

2.4.3 Effective Axial Length

Fringing at the axial ends of the machine extends the effective flux-linking length beyond the physical stack length L_s . A common approximation is [2, Ch. 5]:

$$L_e = L_s + 2g \quad (9)$$

2.4.4 Electrical Loading

The linear current density (electrical loading) characterises the ampere-conductor loading on the stator bore circumference [1]:

$$A_{rms} = \frac{J_{rms} k_{wc} A_{slot}}{\tau_s} \quad (10)$$

where J_{rms} is the rms current density in the conductors, k_{wc} the slot fill factor, and A_{slot} the net slot cross-section.

2.4.5 Electromagnetic Torque – Shear Stress Approach

The electromagnetic shear stress acting at the stator bore is [2, Ch. 3]:

$$\sigma = \frac{1}{2} k_{w,1} B_{g,\text{peak}} A_{\text{peak}} = \frac{1}{\sqrt{2}} k_{w,1} B_{g,\text{peak}} A_{\text{rms}} \quad (11)$$

The resulting average electromagnetic torque and shaft power are:

$$T = \sigma \cdot 2\pi R_{si}^2 L_e, \quad P = T \omega_m \quad (12)$$

2.4.6 Flux Per Pole

The flux per pole, needed to size the back-iron and to calculate the induced EMF, is [2, Ch. 2]:

$$\Phi_{\text{pole}} = B_{g,\text{peak}} \alpha_m \tau_p L_e, \quad \tau_p = \frac{\pi R_{si}}{p} \quad (13)$$

where $\alpha_m = \theta_m/180^\circ$ is the magnet arc ratio ($160^\circ/180^\circ = 0.889$ here) and τ_p is the pole pitch at the stator bore.

2.4.7 Back-Iron Sizing

The stator and rotor yoke widths are set so that the peak flux density in the iron does not exceed the target $B_t = 1.4$ T [2, Ch. 5]:

$$w_{sy} = \frac{\Phi_{\text{pole}}}{2 B_t L_e}, \quad w_{ry} = \frac{\Phi_{\text{pole}}}{2 B_t L_e} \quad (14)$$

3 Q1 – Integral-Slot Winding Design

3.1 Machine Parameters and Notation

The given machine has the following specifications:

Table 1: Given and derived parameters – Q1

Parameter	Meaning	Symbol	Value
Number of stator slots	Total slots around the stator bore	N_s	72
Number of poles	Total magnetic poles on the rotor	$2p$	6
Number of pole pairs	Half the pole count; $p = 2p/2$	p	3
Number of phases	Phases of the AC supply	m	3
Winding layers	Number of coil sides per slot	—	2 (double-layer)

Notation used throughout:

- N_s – total number of stator slots.

- p – number of pole pairs (a 6-pole machine has $p = 3$).
- m – number of phases ($m = 3$).
- α_e – *electrical angle per slot*: the angle (in electrical degrees) between the EMF phasors of two adjacent slots.
- q – *slots per pole per phase*: how many consecutive slots are occupied by one phase under one pole.
- τ_p – *pole pitch* in slots: the number of slots spanned by one pole.
- y – *coil pitch* (or coil span) in slots: the number of slots between the two sides of one coil.
- k_d, k_p, k_w – distribution, pitch, and winding factors, respectively.
- Superscript or subscript n denotes the harmonic order ($n = 1, 3, 5, \dots$).
- A+/A– means Phase A current flows *into* / *out of* the page.

3.2 Step 1 – Derived Quantities

3.2.1 Electrical Angle per Slot α_e

The stator has $N_s = 72$ slots arranged uniformly around the bore. One full revolution is 360° mechanical, but in electrical terms the field repeats once per pole pair, so one full electrical cycle corresponds to $360^\circ/p$ mechanical degrees. The electrical angle between two adjacent slots is therefore:

$$\boxed{\alpha_e = \frac{p \times 360^\circ}{N_s}} = \frac{3 \times 360^\circ}{72} = \mathbf{15^\circ} \text{ (electrical)} \quad (15)$$

3.2.2 Slots per Pole per Phase q

The total number of slots is shared equally among $2p$ poles and m phases:

$$\boxed{q = \frac{N_s}{2p \cdot m}} = \frac{72}{6 \times 3} = \mathbf{4} \quad (16)$$

Because $q = 4 \in \mathbb{Z}^+$, this is an *integral-slot* winding. Each phase occupies 4 consecutive slots under each pole.

3.2.3 Pole Pitch τ_p

The pole pitch is the number of slots under one pole:

$$\boxed{\tau_p = \frac{N_s}{2p}} = \frac{72}{6} = \mathbf{12} \text{ slots} \quad (17)$$

A *full-pitched* coil spans exactly $\tau_p = 12$ slots.

3.2.4 One Pole-Pair = 360° electrical = 24 slots

Since $\alpha_e = 15^\circ$, it takes $360^\circ/15^\circ = 24$ slots to complete one full electrical cycle. This means **one pole-pair spans 24 consecutive slots**, which is the minimum repeating unit of the winding.

3.3 Step 2 – Upper Layer Assignment (the 60° Sector Rule)

3.3.1 Physical Basis

Each stator slot has a conductor whose induced EMF is represented by a phasor. The phasor of slot k (upper layer) points at electrical angle:

$$\varphi_k = (k - 1) \alpha_e \pmod{360^\circ} \quad (18)$$

For a balanced 3-phase winding the 360° circle is divided into **six sectors of 60°** each, alternating between positive and negative directions of the three phases. The sector assignment follows the positive phase sequence A→B→C:

Sector	Angle range	Assignment
1	[0°, 60°)	A+
2	[60°, 120°)	C−
3	[120°, 180°)	B+
4	[180°, 240°)	A−
5	[240°, 300°)	C+
6	[300°, 360°)	B−

Why this ordering? Phase A peaks at 0°. Between A's peak (0°) and B's peak (120°), the phase that is most negative is C (it is at −60° relative to its own peak), so the sector [60°, 120°) is labelled C−. Between B's peak and C's peak the lagging phase A is most negative, hence A− at [180°, 240°), and so on. The result is a sequence that exactly places 4 slots (one group of $q = 4$) in each sector because $4 \times 15^\circ = 60^\circ$ fills one sector precisely.

3.3.2 Slot Angle Table – One Pole-Pair (Slots 1–24)

Table 2: Upper-layer phase assignment for one pole-pair (72-slot, 6-pole)

Slot k	φ_k	Sector	Layer-1	Slot k	φ_k	Sector	Layer-1
1	0°	1	A+	13	180°	4	A−
2	15°	1	A+	14	195°	4	A−
3	30°	1	A+	15	210°	4	A−
4	45°	1	A+	16	225°	4	A−
5	60°	2	C−	17	240°	5	C+
6	75°	2	C−	18	255°	5	C+
7	90°	2	C−	19	270°	5	C+
8	105°	2	C−	20	285°	5	C+
9	120°	3	B+	21	300°	6	B−
10	135°	3	B+	22	315°	6	B−
11	150°	3	B+	23	330°	6	B−
12	165°	3	B+	24	345°	6	B−
Pole 1 (0°–165°)				Pole 2 (180°–345°)			

Note how Pole 2 (slots 13–24) is the *sign-flipped mirror* of Pole 1: $A+ \rightarrow A-$, $C- \rightarrow C+$, $B+ \rightarrow B-$. This is automatic: each slot in Pole 2 is exactly $12 \times 15^\circ = 180^\circ$ ahead of the corresponding slot in Pole 1, placing it in the opposite sector. This 180° phase reversal is what makes the two poles magnetically opposite (one North, one South).

3.4 Step 3 – Lower Layer Assignment

3.4.1 The Coil Concept

A double-layer winding fills each slot with *two* coil sides: Layer 1 (upper) and Layer 2 (lower). One coil consists of a **go side** in the upper layer of slot k and a **return side** in the lower layer of slot $k + y$, where y is the coil pitch (Figure 1).

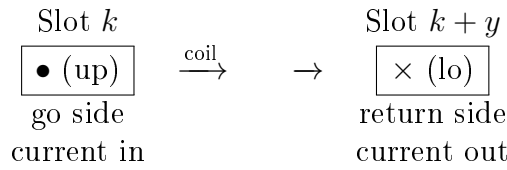


Figure 1: One coil: go side (upper layer, slot k) and return side (lower layer, slot $k + y$).

Therefore, reading it from the perspective of the *lower layer of slot j* :

$$\text{Layer-2 of slot } j = \text{opposite direction of Layer-1 of slot } (j - y) \quad (19)$$

“Opposite direction” means: if Layer-1 of slot $j - y$ is $A+$ (current in), then Layer-2 of slot j is $A-$ (current out), and vice versa.

3.4.2 Full-Pitched Winding ($y = 12$)

Apply rule (19) with $y = 12$. Three representative slots are shown; all 24 follow the same logic:

Slot j	Layer-1 of slot $j - 12$	$\Rightarrow$ Layer-2 of slot j
1	Layer-1[13] = $A-$	$A+$
5	Layer-1[17] = $C+$	$C-$
13	Layer-1[1] = $A+$	$A-$

Key geometric reason: The electrical angle difference between slot j and slot $j - 12$ is $12 \times 15^\circ = 180^\circ$ exactly. Every slot is therefore opposite (in phasor sense) to its partner 12 slots away. Flipping the direction cancels the sign: $-(A-) = A+$. As a result, **both layers in every slot carry the same phase in the same direction:**

Slot:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Layer 1:		A+				C-				B+				A-				C+				B-	
Layer 2:		A+				C-				B+				A-				C+				B-	

The coil pitch arrow: $\underbrace{1 \xrightarrow{y=12} 13}_{\text{Phase A coil 1}}, \underbrace{2 \rightarrow 14}_{\text{coil 2}}, \dots$

3.4.3 11/12 Short-Pitched Winding ($y = 11$)

Now $y = 11$: the angle difference between slot j and slot $j - 11$ is $11 \times 15^\circ = 165^\circ$ (not 180°). At group boundaries the phasor falls in the adjacent sector instead of the opposite one, creating *mixed-phase* slots.

Slot:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Layer 1:			A+				C-				B+				A-		
Layer 2:	A+	A+	A+	C-	C-	C-	C-	B+	B+	B+	B+	A-	A-	A-	A-	C+	C+

The highlighted (yellow) slots (4, 8, 12, 16, 20, 24) are the *boundary slots* where two different phases share the same slot. They arise because shortening the pitch by 1 slot shifts the return conductor out of the purely opposite sector by $180^\circ - 165^\circ = 15^\circ$ (one slot), crossing into the adjacent group.

Worked example for boundary slot $j = 4$:

Layer-1 of slot $j - 11 = \text{slot } -7 \equiv \text{slot } 17$ (wrap by adding 24) = C+.

Layer-2 of slot 4 = $-(C+) = C-$.

But Layer-1 of slot 4 = A+. $\Rightarrow$ **mixed: A+/C-**.

3.5 Step 4 – Distribution Factor k_d

3.5.1 Physical Meaning

In a concentrated winding (all q coils of one phase under one pole placed in a single slot) the coil EMFs are perfectly in phase and add arithmetically. In a *distributed* winding the q coils are spread over q adjacent slots, each displaced by α_e electrically. Their EMF phasors form an arc of a regular polygon, and the vector sum is smaller than the arithmetic sum.

$$k_d = \frac{\text{vector (phasor) sum of the } q \text{ coil EMFs}}{\text{arithmetic sum of the } q \text{ coil EMFs}} \quad (20)$$

3.5.2 Derivation

Each of the q phasors has magnitude E_m and successive phasors are offset by α_e . The vector sum of q unit phasors starting at 0° and stepping by α_e :

$$\left| \sum_{i=0}^{q-1} e^{j i \alpha_e} \right| = \frac{\sin(q \alpha_e / 2)}{\sin(\alpha_e / 2)} \quad (21)$$

(standard result for a finite geometric series on the unit circle). The arithmetic sum is simply q . Hence:

$$k_{d,n} = \frac{\sin\left(\frac{n q \alpha_e}{2}\right)}{q \sin\left(\frac{n \alpha_e}{2}\right)} \quad (22)$$

where n is the harmonic order. For the n -th harmonic the electrical angle between adjacent phasors becomes $n \alpha_e$, so α_e is replaced by $n \alpha_e$ throughout.

3.5.3 Calculations for $q = 4$, $\alpha_e = 15^\circ$

Fundamental ($n = 1$):

$$k_{d,1} = \frac{\sin\left(\frac{1 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{1 \times 15^\circ}{2}\right)} = \frac{\sin 30^\circ}{4 \sin 7.5^\circ} = \frac{0.5000}{4 \times 0.1305} = \frac{0.5000}{0.5221} = \mathbf{0.9577} \quad (23)$$

Third harmonic ($n = 3$):

$$k_{d,3} = \frac{\sin\left(\frac{3 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{3 \times 15^\circ}{2}\right)} = \frac{\sin 90^\circ}{4 \sin 22.5^\circ} = \frac{1.0000}{4 \times 0.3827} = \frac{1.0000}{1.5307} = \mathbf{0.6533} \quad (24)$$

Fifth harmonic ($n = 5$):

$$k_{d,5} = \frac{\sin\left(\frac{5 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{5 \times 15^\circ}{2}\right)} = \frac{\sin 150^\circ}{4 \sin 37.5^\circ} = \frac{0.5000}{4 \times 0.6088} = \frac{0.5000}{2.4351} = \mathbf{0.2053} \quad (25)$$

The distribution factor *always decreases with harmonic order* for a fixed q – distributing the winding inherently attenuates higher harmonics.

3.6 Step 5 – Pitch Factor k_p

3.6.1 Physical Meaning

Consider one coil with its go side in slot k and its return side in slot $k + y$. The EMF induced in the go-side conductor is $e_{go}(t) = E_m \sin(\omega t)$. The EMF induced in the return-side conductor is $e_{ret}(t) = E_m \sin(\omega t - \theta_c)$ where $\theta_c = y \alpha_e$ is the electrical angle spanned by the coil. Because the return conductor is wound backwards (it is the return path of the coil), its contribution to the *net* coil EMF is $-e_{ret}$. The total coil EMF is:

$$e_{coil} = e_{go} - e_{ret} = E_m [\sin \omega t - \sin(\omega t - \theta_c)] = 2E_m \cos\left(\omega t - \frac{\theta_c}{2}\right) \sin \frac{\theta_c}{2} \quad (26)$$

The amplitude is $2E_m \sin(\theta_c/2)$. For a *full-pitched* coil $\theta_c = 180^\circ$, so the amplitude is $2E_m \sin(90^\circ) = 2E_m$ (maximum possible). The *pitch factor* is the ratio of the short-pitch amplitude to the full-pitch amplitude:

$$k_p = \frac{2E_m \sin(\theta_c/2)}{2E_m \sin(90^\circ)} = \sin \frac{\theta_c}{2} = \sin \frac{y \alpha_e}{2} = \sin \frac{y \pi}{2\tau_p} \quad (27)$$

For the n -th harmonic the field rotates n times faster, so the coil spans n times more electrical angle:

$$k_{p,n} = \sin \left(\frac{n y \pi}{2 \tau_p} \right) \quad (28)$$

3.6.2 Calculations

Full-pitched ($y = 12, \tau_p = 12$):

$$k_{p,1} = \sin \frac{1 \times 12 \pi}{2 \times 12} = \sin \frac{\pi}{2} = \mathbf{1.0000} \quad (29)$$

$$k_{p,3} = \left| \sin \frac{3 \times 12 \pi}{2 \times 12} \right| = \left| \sin \frac{3\pi}{2} \right| = |-1| = \mathbf{1.0000} \quad (30)$$

$$k_{p,5} = \left| \sin \frac{5 \times 12 \pi}{2 \times 12} \right| = \left| \sin \frac{5\pi}{2} \right| = |+1| = \mathbf{1.0000} \quad (31)$$

All harmonics have $k_p = 1$ – a full-pitched coil gives maximum EMF for every harmonic.

11/12 Short-pitched ($y = 11, \tau_p = 12$):

$$k_{p,1} = \sin \frac{1 \times 11 \pi}{2 \times 12} = \sin \frac{11\pi}{24} = \sin 82.5^\circ = \mathbf{0.9914} \quad (32)$$

$$k_{p,3} = \left| \sin \frac{3 \times 11 \pi}{2 \times 12} \right| = \left| \sin \frac{33\pi}{24} \right| = |\sin 247.5^\circ| = |-\sin 67.5^\circ| = \mathbf{0.9239} \quad (33)$$

$$k_{p,5} = \left| \sin \frac{5 \times 11 \pi}{2 \times 12} \right| = \left| \sin \frac{55\pi}{24} \right| = |\sin 412.5^\circ| = \sin 52.5^\circ = \mathbf{0.7934} \quad (34)$$

3.7 Step 6 – Winding Factor k_w

The winding factor combines both effects:

$$k_{w,n} = k_{d,n} \times k_{p,n} \quad (35)$$

Table 3: Complete winding factor results – Full-pitched vs. 11/12 short-pitched

n	Full-pitched ($y = 12$)			11/12 Short-pitched ($y = 11$)			Change	
	$k_{d,n}$	$k_{p,n}$	$k_{w,n}$	$k_{d,n}$	$k_{p,n}$	$k_{w,n}$	Δk_w	$\Delta\%$
1 (fund.)	0.9577	1.0000	0.9577	0.9577	0.9914	0.9495	-0.0082	-0.9%
3	0.6533	1.0000	0.6533	0.6533	0.9239	0.6036	-0.0497	-7.6%
5	0.2053	1.0000	0.2053	0.2053	0.7934	0.1629	-0.0424	-20.6%

3.8 Step 7 – Winding Diagram (One Pole-Pair)

Table 4 shows the complete double-layer slot layout for one pole-pair (slots 1–24). The symbol $\bullet$ means current flows *into* the page; $\times$ means current flows *out of* the page. The coil pitch arrow $k \rightarrow k + y$ connects each go-side (Layer 1) to its return-side (Layer 2).

Table 4: Winding diagram – full-pitched ($y = 12$), one pole-pair

Slot k	Pole 1												Pole 2											
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	
Layer 1		A●				C×				B●				A×				C●					B×	
Layer 2		A●				C×				B●				A×				C●					B×	
Coil span	$\longleftarrow y = 12 \text{ slots} \longrightarrow$																							

Table 5: Winding diagram – 11/12 short-pitched ($y = 11$), one pole-pair. Yellow cells = boundary slots with mixed-phase layers.

Slot k	Pole 1												Pole 2				
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Layer 1			A $\bullet$				C $\times$				B $\bullet$			A $\times$			
Layer 2	A $\bullet$	A $\bullet$	A $\bullet$	C $\times$	C $\times$	C $\times$	C $\times$	B $\bullet$	B $\bullet$	B $\bullet$	B $\bullet$	A $\times$	A $\times$	A $\times$	A $\times$	C $\bullet$	C $\bullet$
Coil span	$\leftarrow y = 11 \text{ slots} \rightarrow$																

3.9 Discussion and Comments

1. **Distribution factor decreases with harmonic order.** For $q = 4$ and $\alpha_e = 15^\circ$, going from $n = 1$ to $n = 5$ drops k_d from 0.958 to only 0.205. Spreading the winding over q slots is therefore a powerful tool for harmonic attenuation, even before any pitch adjustment.
2. **Full-pitched winding: pitch factor is unity for all odd harmonics.** Because $y\alpha_e = 180^\circ$, the formula $k_{p,n} = \sin(n\pi/2)$ gives ± 1 for every odd n . This means the full-pitched winding provides *no* pitch-based harmonic suppression – the 3rd harmonic winding factor of 0.653 is entirely due to k_d and remains high.
3. **Short-pitching disproportionately suppresses higher harmonics.** Shortening by just 1 slot ($y = 11$) changes $k_{p,1}$ by only -0.9% but reduces $k_{p,5}$ by -20.6% . The mathematical reason: $k_{p,n} = \sin(n \cdot y\pi/2\tau_p)$ is more sensitive to changes in y for larger n because the argument is multiplied by n . A small shift in y sweeps a larger angle for the 5th harmonic than for the fundamental.
4. **The 3rd harmonic is a triplen harmonic.** In a balanced 3-phase machine, triplen ($n = 3, 9, 15, \dots$) space harmonics do not produce average torque because they are in phase across all three phases (no 120° offset between phases for $n = 3$). However, they still cause iron losses and noise.

5. **The 5th harmonic produces a *backward-rotating* field.** The 5th space harmonic rotates at $1/5$ of the fundamental speed but in the *opposite* direction. It contributes a parasitic braking torque and is a primary source of torque ripple at 6th-order frequency (5th + 7th harmonics together create the 6th harmonic torque pulsation). Reducing $k_{w,5}$ from 0.205 to 0.163 therefore meaningfully reduces this pulsation.
6. **Summary recommendation:** The 11/12 short-pitched winding is the better design. It costs $< 1\%$ of useful torque capability while achieving significant attenuation of harmful harmonics. In industrial practice, short pitching is almost universally applied in distributed-winding AC machines [1, 3].

4 Q2 – Fractional-Slot Winding Design

4.1 Machine Parameters and Notation

Table 6: Design 1 parameters ($2575421 \bmod 5 = 1$)

Parameter	Symbol	Value
Number of stator slots	N_s	15
Number of poles (magnets)	N_m	8
Number of pole pairs	p	4
Phases	m	3
R_{ro}/R_{so} ratio	—	0.6

The slots per pole per phase is:

$$q = \frac{N_s}{N_m \cdot m} = \frac{15}{8 \times 3} = 0.625 \notin \mathbb{Z}^+ \quad (36)$$

Because q is not an integer, the simple coil-group rule of integral-slot windings (“ q consecutive slots per phase per pole”) does not apply, and the closed-form formulas for k_d and k_p are invalid. Instead, the *star of slots* phasor method must be used.

4.2 Step 1 – Electrical Angle per Slot α_s

For a machine with N_m poles the field repeats every pole pair (360° electrical). The electrical angle between two adjacent slots is:

$$\boxed{\alpha_s = \frac{N_m \times 180^\circ}{N_s}} = \frac{8 \times 180^\circ}{15} = \mathbf{96^\circ} \quad (37)$$

Note that $\alpha_s = 96^\circ > 60^\circ$: each step jumps *across* a sector boundary rather than staying within one 60° phase group. This is the signature of a fractional-slot winding – the slot phasors do not cluster neatly into phase groups.

4.3 Step 2 – Slot Phasor Angles (Star of Slots)

The EMF phasor of slot k points at electrical angle (Eq. (5)):

$$\varphi_k = (k - 1) \alpha_s \pmod{360^\circ} \quad (38)$$

Table 7 lists the calculation for all 15 slots. Because $\gcd(15, 4) = 1$, no two slots share the same angle – all 15 phasors are distinct and the sequence visits 15 different positions before returning to 0° .

Table 7: Slot phasor angles – Design 1 ($\alpha_s = 96^\circ$)

Slot k	$(k-1) \times 96^\circ$	φ_k	Sector	Slot k	$(k-1) \times 96^\circ$	φ_k	Sector
1	0	0°	1	9	768	48°	1
2	96	96°	2	10	864	144°	3
3	192	192°	4	11	960	240°	5
4	288	288°	5	12	1056	336°	6
5	384	24°	1	13	1152	72°	2
6	480	120°	3	14	1248	168°	3
7	576	216°	4	15	1344	264°	5
8	672	312°	6				

4.4 Step 3 – Phase Assignment (60° Sector Rule)

The same six-sector rule from Q1 (Section 3) applies here. The 360° circle is divided into six 60° sectors, and each slot is assigned to the phase whose sector contains its phasor:

Sector	Angle range	Assignment
1	$[0^\circ, 60^\circ)$	A+
2	$[60^\circ, 120^\circ)$	C–
3	$[120^\circ, 180^\circ)$	B+
4	$[180^\circ, 240^\circ)$	A–
5	$[240^\circ, 300^\circ)$	C+
6	$[300^\circ, 360^\circ)$	B–

Applying this to Table 7 gives the full per-slot assignment in Table 8.

Table 8: Complete phase assignment – all 15 slots

Slot	φ_k	Sector	Assignment	Slot	φ_k	Sector	Assignment
1	0°	1	A+	9	48°	1	A+
2	96°	2	C–	10	144°	3	B+
3	192°	4	A–	11	240°	5	C+
4	288°	5	C+	12	336°	6	B–
5	24°	1	A+	13	72°	2	C–
6	120°	3	B+	14	168°	3	B+
7	216°	4	A–	15	264°	5	C+
8	312°	6	B–				

The result is balanced: each phase receives exactly $15/3 = 5$ slots.

- **Phase A:** A+ at slots 1, 5, 9 and A– at slots 3, 7
- **Phase B:** B+ at slots 6, 10, 14 and B– at slots 8, 12
- **Phase C:** C+ at slots 4, 11, 15 and C– at slots 2, 13

Figure 2 shows all 15 slot phasors arranged in the star of slots, colour-coded by phase.

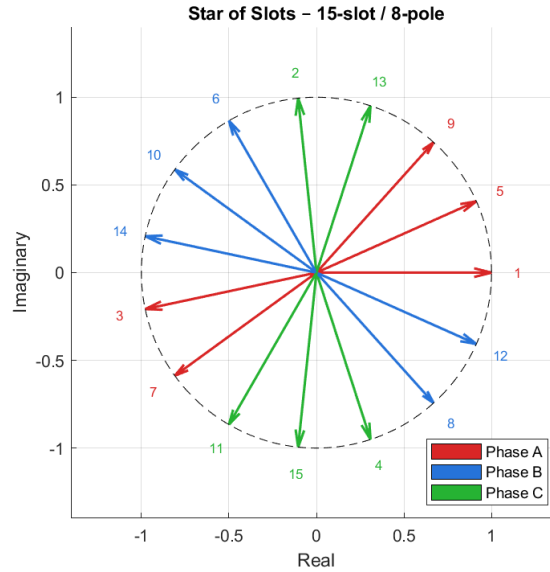


Figure 2: Star of slots for Design 1 ($N_s = 15$, $N_m = 8$). Colours: red = Phase A, blue = Phase B, green = Phase C. Solid arrows = positive direction; dashed = negative direction.

4.5 Step 4 – Winding Factor by Phasor Summation

4.5.1 Calculation Method

For the n -th space harmonic, each slot assigned to Phase A contributes one unit phasor. The angle of that phasor depends on whether the slot is positive or negative (Eq. (6)):

$$\text{A+ slot } k: e^{jn\varphi_k}, \quad \text{A- slot } k: e^{j(n\varphi_k + 180^\circ)} \quad (39)$$

The winding factor is then the magnitude of the vector sum divided by the number of slots per phase ($N_{ph} = 5$):

$$k_{w,n} = \frac{|\Sigma_A|}{N_{ph}} = \frac{|\Sigma_A|}{5} \quad (40)$$

The A– slots effectively have their phasors *reversed* (rotated 180°) before summing, which accounts for the fact that the current flows in the opposite direction in those slots.

4.5.2 Fundamental ($n = 1$)

Table 9: Phase A phasor contributions, $n = 1$

Slot	Type	φ_k	Angle used ($n\varphi_k + \text{rev.}$)	Real	Imag.
1	A+	0°	0°	+1.000	+0.000
5	A+	24°	24°	+0.914	+0.407
9	A+	48°	48°	+0.669	+0.743
3	A–	192°	$192^\circ + 180^\circ = 12^\circ$	+0.978	+0.208
7	A–	216°	$216^\circ + 180^\circ = 36^\circ$	+0.809	+0.588
Sum Σ_A				+4.370	+1.946

$$|\Sigma_A| = \sqrt{4.370^2 + 1.946^2} = \sqrt{19.10 + 3.79} = 4.783 \quad (41)$$

$$k_{w,1} = \frac{4.783}{5} = 0.9566 \quad (42)$$

Observation: After reversing the A– phasors (slots 3 and 7), all five phasors cluster in the first quadrant within a 48° arc. They nearly align, which explains the high winding factor.

4.5.3 Third Harmonic ($n = 3$)

For $n = 3$ the phasor angle of each slot is multiplied by 3 before applying the reversal for A– slots.

Table 10: Phase A phasor contributions, $n = 3$

Slot	Type	φ_k	Angle used ($3\varphi_k + \text{rev.}$)	Real	Imag.
1	A+	0°	0°	+1.000	+0.000
5	A+	24°	72°	+0.309	+0.951
9	A+	48°	144°	–0.809	+0.588
3	A–	192°	$576^\circ + 180^\circ = 36^\circ$	+0.809	+0.588
7	A–	216°	$648^\circ + 180^\circ = 108^\circ$	–0.309	+0.951
Sum Σ_A				+1.000	+3.078

$$|\Sigma_A| = \sqrt{1.000^2 + 3.078^2} = \sqrt{1.000 + 9.474} = 3.236 \quad (43)$$

$$k_{w,3} = \frac{3.236}{5} = 0.6472 \quad (44)$$

4.5.4 Fifth Harmonic ($n = 5$)

Table 11: Phase A phasor contributions, $n = 5$

Slot	Type	φ_k	Angle used ($5\varphi_k + \text{rev.}$)	Real	Imag.
1	A+	0°	0°	+1.000	+0.000
5	A+	24°	120°	−0.500	+0.866
9	A+	48°	240°	−0.500	−0.866
3	A−	192°	$960^\circ + 180^\circ = 60^\circ$	+0.500	+0.866
7	A−	216°	$1080^\circ + 180^\circ = 180^\circ$	−1.000	+0.000
Sum Σ_A				−0.500	+0.866

$$|\Sigma_A| = \sqrt{(-0.500)^2 + (0.866)^2} = \sqrt{0.250 + 0.750} = 1.000 \quad (45)$$

$$k_{w,5} = \frac{1.000}{5} = 0.200 \quad (46)$$

Observation: The three A+ slots lie at 0° , 120° , 240° – a balanced triphasic set whose vector sum is exactly zero. The A− slots only partially restore the sum, resulting in a very small $|\Sigma_A| = 1.0$, hence the low $k_{w,5}$.

Table 12: Summary of winding factors – Design 1 (15-slot/8-pole)

Harmonic	Σ_A	$ \Sigma_A $	$k_{w,n}$	Note
$n = 1$ (fund.)	$4.370 + j1.946$	4.783	0.9566	High; good torque density
$n = 3$ (triplen)	$1.000 + j3.078$	3.236	0.6472	Zero-seq.; no net torque
$n = 5$	$-0.500 + j0.866$	1.000	0.2000	Naturally suppressed

Figure 3 shows the Phase A phasor diagrams for all three harmonics. Each arrow is one slot phasor (after reversing A− slots); the thick red arrow is the vector resultant.

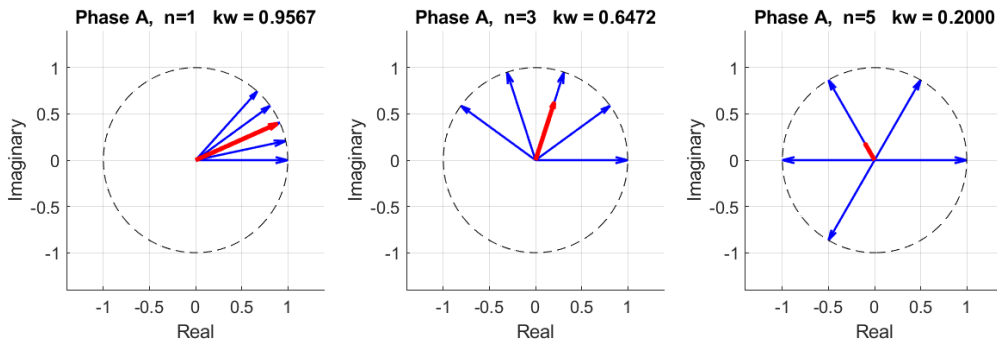


Figure 3: Phase A phasor diagrams for $n = 1, 3, 5$. Blue arrows = individual slot phasors; red arrow = vector sum. Winding factor = $|\text{sum}|/5$.

Note on the Concentrated-Coil Convention

The phasor sum above uses only the five *go-side* positions of Phase A (Layer 1: slots 1, 5, 9, 3, 7). The return-sides (Layer 2) of these coils are *not* added as extra phasors – and this is intentional.

In a fractional-slot **concentrated** (single-tooth) winding each coil is wound around *one* tooth. Following the convention used by Hanselman [2], each coil is represented by a *single phasor placed at its tooth-centre angle*, and the winding factor is the vector sum of the five coil phasors divided by 5:

$$k_{w,n} = \frac{|\sum_{i=1}^5 e^{jn\varphi_i}|}{5} \quad (\text{coil phasors at tooth centres}). \quad (47)$$

The return-side is the same physical conductor as the go-side – both together form a single coil – so adding it as a second phasor would be double counting.

Why some software (e.g. Emetor) reports a different value. If one instead treats every conductor side as an independent slot phasor and normalises by 10 (5 go-sides + 5 return-sides), a further pitch factor $k_p = \sin(\alpha_s/2) = \sin 48^\circ = 0.743$ enters, giving $0.9566 \times 0.743 \approx 0.71$. That number describes the same winding – it just hides the pitch factor inside the sum rather than declaring it separately. For the design equations used in this report (EMF, torque, shear stress, turns/phase, §5) the correct quantity is the concentrated-coil value $k_{w,1} = 0.9566$.

4.6 Step 5 – Constructing the Winding Diagram

4.6.1 From Slot Assignments to Physical Coils

The star-of-slots gave us *one assignment per slot* – the *upper (Layer 1)* of each slot. In the physical machine each slot has **two** coil-side layers, and each coil is wound around **one tooth** (concentrated / tooth-coil winding).

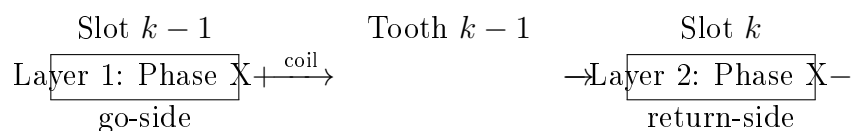
The double-layer layout follows two rules:

1. **Layer 1 of slot k** = star-of-slots assignment (Table 8) – the *go-side* of the coil on tooth k .
2. **Layer 2 of slot k** = *opposite* of Layer 1 of slot $k - 1$ (circular: slot 0 $\equiv$ slot 15):

$$\text{Layer-2}[k] = -\text{Layer-1}[k - 1] \quad (48)$$

This is the *return-side* of the coil on tooth $k - 1$.

Physical meaning of Eq. (48): Tooth $k - 1$ sits between slot $k - 1$ and slot k . A coil wound around tooth $k - 1$ has its go-side in Layer 1 of slot $k - 1$ and its return-side in Layer 2 of slot k . The return side always carries the *opposite* current direction.



4.6.2 Worked Example (Slot 4)

Layer-1[4] = C+ (from Table 8).

Layer-2[4] = −Layer-1[3] = −(A−) = A+.

This means slot 4 has C+ in the upper layer and A+ in the lower layer. These belong to *different* phases – a normal feature of double-layer FSCWs.

4.6.3 Coil Identification Table

Table 13 shows all 15 coils (one per tooth). The go-side is always Layer 1 of slot k ; the return-side is always Layer 2 of slot $k + 1$ (which is the opposite of Layer 1 of slot k , confirming every coil is single-phase).

Table 13: Coil table – 15 tooth-wound coils

Tooth	Go-side: Layer-1[slot k]	Return-side: Layer-2[slot $k+1$]	Phase
1	Slot 1: A+	Slot 2: A−	A
2	Slot 2: C−	Slot 3: C+	C
3	Slot 3: A−	Slot 4: A+	A
4	Slot 4: C+	Slot 5: C−	C
5	Slot 5: A+	Slot 6: A−	A
6	Slot 6: B+	Slot 7: B−	B
7	Slot 7: A−	Slot 8: A+	A
8	Slot 8: B−	Slot 9: B+	B
9	Slot 9: A+	Slot 10: A−	A
10	Slot 10: B+	Slot 11: B−	B
11	Slot 11: C+	Slot 12: C−	C
12	Slot 12: B−	Slot 13: B+	B
13	Slot 13: C−	Slot 14: C+	C
14	Slot 14: B+	Slot 15: B−	B
15	Slot 15: C+	Slot 1: C−	C

Coil distribution: **Phase A** – teeth 1, 3, 5, 7, 9; **Phase B** – teeth 6, 8, 10, 12, 14; **Phase C** – teeth 2, 4, 11, 13, 15.

Note the grouping pattern: Phase A occupies five consecutive odd-numbered teeth (1–9), Phase B five alternating teeth in the upper half (6–14), and Phase C the remaining five teeth. This non-uniform spacing (unlike the regular ABC pattern of integral-slot windings) is a direct consequence of the fractional-slot geometry.

4.6.4 Double-Layer Slot Table

Combining Layer 1 (from Table 8) and Layer 2 (from Eq. (48)) gives the full double-layer layout:

Table 14: Double-layer slot layout – Design 1 (15-slot/8-pole)

Slot	Layer 1 (Upper)	Layer 2 (Lower)	Slot	Layer 1 (Upper)	Layer 2 (Lower)
1	A+	C−	9	A+	B+
2	C−	A−	10	B+	A−
3	A−	C+	11	C+	B−
4	C+	A+	12	B−	C−
5	A+	C−	13	C−	B+
6	B+	A−	14	B+	C+
7	A−	B−	15	C+	B−
8	B−	A+			

How to read the table: Each row is one slot. Layer 1 (upper, closer to the air gap) is always the go-side of one coil. Layer 2 (lower) is always the return-side of the adjacent coil (on the preceding tooth). In most slots two *different* phases share the same slot (e.g., slot 1 has A+ above and C− below) – this is entirely normal for a double-layer FSCW.

The full winding layout is shown in Figure 4. Each arc in the figure represents one tooth-wound coil connecting the two adjacent slots of that tooth.

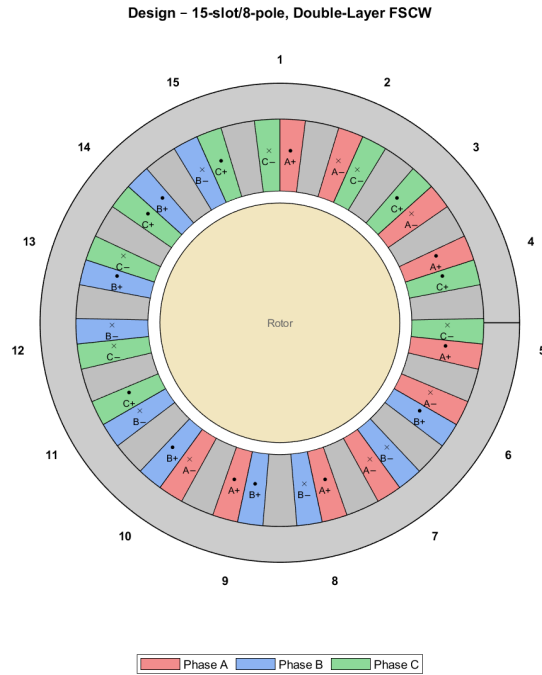


Figure 4: Winding layout for Design 1 (15-slot/8-pole, 3-phase). Each arc = one tooth-wound coil. Red = Phase A (teeth 1,3,5,7,9), blue = Phase B (teeth 6,8,10,12,14), green = Phase C (teeth 2,4,11,13,15). ● = current into page; × = current out of page.

4.7 Step 6 – Verification: Valid and Balanced Winding

A winding is *valid* if every slot is occupied and every coil belongs to exactly one phase. A winding is *balanced* if all three phases produce equal RMS EMFs displaced by 120° .

1. **All slots occupied.** Table 14 fills all $15 \times 2 = 30$ conductor positions with no gaps.
2. **Every coil is single-phase.** Table 13 shows that the go-side and return-side of every coil carry the same phase letter (only the $\pm$ sign differs, as required for a coil). No coil mixes phases.
3. **Equal conductors per phase.** Each phase has 5 coils = 10 conductor positions out of 30 total (= $1/3$) – identical for A, B, and C.
4. **Balanced phase resultants.** By the three-fold symmetry of the winding ($\gcd(15, 3) = 3$), the phasor resultants of phases B and C have the same magnitude as Phase A ($|\Sigma_B| = |\Sigma_C| = 4.783$ for $n = 1$) and are displaced by exactly 120° .
5. **Winding factor check.** $k_{w,1} = 0.9566$ is consistent with the value tabulated in [2, Table] for the 15/8 slot-pole combination, confirming the phasor summation is correct.

4.8 Discussion

1. **High fundamental winding factor for a concentrated winding.** $k_{w,1} = 0.9566$ is comparable to that of distributed windings (typically 0.925–0.966) and far above the minimum (≈ 0.5) for random concentrated windings. The 15/8 combination is specifically identified in [2] as one of the best fractional-slot combinations for torque density.
2. **Natural suppression of the 5th harmonic.** The three A+ slots contribute phasors at 0° , 120° , 240° for $n = 5$ – a balanced triphasic set that sums to zero. Only the A– slots restore a small residual ($|\Sigma| = 1.0$), giving $k_{w,5} = 0.2$. This suppression is geometric and requires no pitch adjustment.
3. **3rd harmonic: large winding factor, zero net rotating field.** $k_{w,3} = 0.647$ is large, but as discussed in Q1, the 3rd space harmonic is triplen: all three phases produce it in the same spatial direction, and the balanced condition $I_A + I_B + I_C = 0$ cancels the net rotating component. It may still cause iron losses through standing-wave effects.
4. **Two phases share each slot.** Unlike integral-slot windings where each slot is occupied by a single phase, the double-layer FSCW places two *different* phases in each slot. This is unavoidable when $q < 1$: adjacent teeth belong to different phases, and each slot borders two adjacent teeth.
5. **Short end-windings – key FSCW advantage.** Because each coil spans only one tooth pitch (approximately $360^\circ/(N_s) = 24^\circ$ mechanical), the overhang is very short. This reduces copper volume, DC resistance, and copper losses – the primary advantage of FSCWs for direct-drive, high-torque-density applications [4].

5 Q3 – Analytical Modelling

5.1 Machine Dimensions

From the assignment common specifications and the Design 1 ratio $R_{ro}/R_{so} = 0.6$:

Table 15: Machine geometry – Design 1

Parameter	Symbol	Value
Stator outer radius	R_{so}	50.0 mm
Rotor outer radius	$R_{ro} = 0.6 R_{so}$	30.0 mm
Air-gap length	g	1.0 mm
Magnet radial thickness	l_m	4.0 mm
Stator inner radius	$R_{si} = R_{ro} + g$	31.0 mm
Rotor core outer radius	$R_{ro,c} = R_{ro} - l_m$	26.0 mm
Stack length	L_s	100 mm
Magnet remanence	B_r	1.3 T
Magnet relative permeability	μ_r	1.05
Magnet arc ratio	α_m	0.889
Target iron flux density	B_t	1.4 T
Current density (rms)	J_{rms}	5 A/mm ²
Slot fill factor	k_{wc}	0.60

5.2 Slot Pitch and Tooth Geometry

The slot pitch at the stator bore:

$$\tau_s = \frac{2\pi R_{si}}{N_s} = \frac{2\pi \times 31}{15} = 12.99 \text{ mm} \quad (49)$$

A first-pass estimate of the peak air-gap flux density without Carter correction gives $B_{g,0} = B_r l_m / (l_m + \mu_r g) = 1.030 \text{ T}$. The average flux density under the magnet arc is $B_{g,avg} = B_{g,0} \alpha_m = 1.030 \times 0.889 = 0.916 \text{ T}$. For $B_t = 1.4 \text{ T}$ in the teeth:

$$w_{tb} = \frac{B_{g,avg} \tau_s}{B_t} = \frac{0.916 \times 12.99}{1.4} = 8.50 \text{ mm} \quad (50)$$

$$w_s = \tau_s - w_{tb} = 12.99 - 8.50 = 4.49 \text{ mm} \quad (51)$$

5.3 Carter's Coefficient

Applying Eq. (8) with $g = 1 \text{ mm}$ and $w_s = 4.49 \text{ mm}$:

$$x = \frac{w_s}{2g} = \frac{4.49}{2} = 2.245 \quad (52)$$

$$\gamma = \frac{4 \times 1}{\pi} [2.245 \arctan(2.245) - \ln \sqrt{1 + 2.245^2}] = \frac{4}{\pi} [2.245 \times 1.150 - 0.899] = 2.143 \text{ mm} \quad (53)$$

$$k_c = \frac{12.99}{12.99 - 2.143} = \mathbf{1.197} \quad (54)$$

5.4 Effective Air Gap and Corrected Flux Density

$$g_{\text{eff}} = k_c g + \frac{l_m}{\mu_r} = 1.197 \times 1 + \frac{4}{1.05} = 1.197 + 3.810 = 5.007 \text{ mm} \quad (55)$$

Substituting into Eq. (7):

$$B_{g,\text{peak}} = \frac{1.3 \times 4}{4 + 1.05 \times 1.197 \times 1} = \frac{5.2}{5.257} = \mathbf{0.989 \text{ T}} \quad (56)$$

5.5 Effective Axial Length

From Eq. (9):

$$L_e = 100 + 2 \times 1 = \mathbf{102 \text{ mm}} \quad (57)$$

5.6 Air-Gap Peak Flux Density Summary

All three values are consistent: the slotting correction reduces $B_{g,\text{peak}}$ from 1.030 T (idealised) to 0.989 T (with Carter correction), a reduction of 3.9%.

5.7 Flux Per Pole

From Eq. (13):

$$\tau_p = \frac{\pi \times 31}{4} = 24.35 \text{ mm} \quad (58)$$

$$\Phi_{\text{pole}} = 0.989 \times 0.889 \times 24.35 \times 10^{-3} \times 102 \times 10^{-3} = \mathbf{2.18 \text{ mWb}} \quad (59)$$

5.8 Magnetic Loading

$$\bar{B} = B_{g,\text{peak}} \cdot \alpha_m = 0.989 \times 0.889 = \mathbf{0.879 \text{ T}} \quad (60)$$

5.9 Iron Yoke Widths

From Eq. (14):

$$w_{sy} = w_{ry} = \frac{2.18 \times 10^{-3}}{2 \times 1.4 \times 0.102} = \mathbf{7.64 \text{ mm}} \quad (61)$$

The rotor shaft radius is $R_{ro,c} - w_{ry} = 26 - 7.64 = 18.4 \text{ mm}$.

5.10 Electrical Loading

Available slot height and slot area (rectangular approximation):

$$h_{\text{slot}} = (R_{so} - R_{si}) - w_{sy} = 19 - 7.64 = 11.36 \text{ mm} \quad (62)$$

$$A_{\text{slot}} = w_s \times h_{\text{slot}} = 4.49 \times 11.36 = 51.0 \text{ mm}^2 \quad (63)$$

From Eq. (10):

$$A_{rms} = \frac{5 \times 0.60 \times 51.0}{12.99} = \mathbf{11.78 \text{ kA/m}} \quad (64)$$

5.11 Shear Stress and Expected Torque

From Eqs. (11)–(12):

$$\sigma = \frac{1}{\sqrt{2}} \times 0.9566 \times 0.989 \times 11,780 = \mathbf{7.87 \text{ kPa}} \quad (65)$$

$$T = 7870 \times 2\pi \times (0.031)^2 \times 0.102 = \mathbf{4.85 \text{ N} \cdot \text{m}} \quad (66)$$

$$P = 4.85 \times \frac{1500 \times 2\pi}{60} = \mathbf{762 \text{ W}} \quad (67)$$

The shear stress of 7.87 kPa is consistent with the range 5–35 kPa quoted for liquid-cooled PMSMs in [2, Ch. 3].

5.12 Number of Turns Per Phase

For $V_{dc} = 48 \text{ V}$ (star connection), the peak phase voltage is:

$$\hat{V}_{ph} = \frac{V_{dc}}{\sqrt{3}} = \frac{48}{\sqrt{3}} = 27.7 \text{ V} \quad (68)$$

The electrical angular frequency at $n = 1500 \text{ rpm}$:

$$\omega_e = p\omega_m = 4 \times \frac{1500 \times 2\pi}{60} = 628.3 \text{ rad/s} \quad (69)$$

The required number of turns per phase [2, Ch. 2]:

$$N_{ph} = \frac{\hat{V}_{ph}}{k_{w,1} \omega_e \Phi_{pole}} = \frac{27.7}{0.9566 \times 628.3 \times 2.18 \times 10^{-3}} \approx 21.2 \quad (70)$$

Rounding to an integer number of conductors per slot ($N_{cs} = 2N_{ph}m/N_s$, must be integer):

$$N_{cs} = 8 \implies N_{ph} = \frac{8 \times 15}{2 \times 3} = 20 \quad (71)$$

Back-EMF at 1500 rpm: $\hat{E}_{ph} = 0.9566 \times 628.3 \times 2.18 \times 10^{-3} \times 20 = 26.2 \text{ V}$

5.13 Summary of Analytical Results

Table 16: Analytical modelling results – Design 1

Parameter	Symbol	Value
Slot pitch	τ_s	12.99 mm
Tooth width	w_{tb}	8.50 mm
Slot opening	w_s	4.49 mm
Carter coefficient	k_c	1.197
Effective air gap	g_{eff}	5.007 mm
Effective axial length	L_e	102 mm
Air-gap peak flux density	$B_{g,\text{peak}}$	0.989 T
Flux per pole	Φ_{pole}	2.18 mWb
Magnetic loading	$\bar{B}$	0.879 T
Stator yoke width	w_{sy}	7.64 mm
Rotor yoke width	w_{ry}	7.64 mm
Slot height	h_{slot}	11.36 mm
Electrical loading (rms)	A_{rms}	11.78 kA/m
Shear stress	σ	7.87 kPa
Expected torque	T	4.85 N·m
Expected output power	P	762 W
Turns per phase	N_{ph}	20
Conductors per slot	N_{cs}	8

6 Q4 – 2D FEA Modelling (ANSYS Maxwell)

(To be completed after ANSYS Maxwell simulation.)

The following results will be obtained from the 2D magnetostatic/transient simulation in ANSYS Maxwell:

1. Flux density distribution in the stator core, with maximum values verified against $B_t = 1.4$ T in teeth and back-iron.
2. Air-gap flux density at the mid-gap radius plotted vs. circumferential angle.
3. Leakage flux path visualisation (slot leakage, magnet-to-magnet, tip leakage, magnet-tooth-magnet).
4. Flux per pole from FEA (to be compared with $\Phi_{\text{pole}}^{\text{analytical}} = 2.18$ mWb).
5. (Bonus) Cogging torque vs. rotor angle.

References

References

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A MATLAB Script

The complete MATLAB script `ee568_hw2_analytical.m` is submitted alongside this report. It computes all Q1–Q3 results, generates the winding diagrams and phasor plots in Section 4 (Figures 2–4), and prints a summary table to the console.